GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting

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Certificate

This is to certify that the report entitled "GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting" submitted by Pedapatruni Shanmuk Vardhan (B123089), Shaktimayee Mohanty (B123118) and Yuvraj Singh (B123153) to International Institute of Information Technology, Bhubaneswar is a record of bonafide project work under my supervision, and the report is submitted for the evaluation of the BTech Degree Final-Semester Project.

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Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei*. It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. The published machine learning literature on this disease is confined to laboratory identification of the organism by spectroscopic methods, and no population-level geographic risk model has been reported.

This report presents GlandersWatch, a platform built from three independently deployable components: a calibrated machine learning risk engine, a retrieval-augmented layer that explains each prediction with citations to official guidance, and an alert service that delivers the result to equine owners. The decomposition is the contribution as much as the model is. Each component communicates through a versioned schema rather than shared internals, none hardcodes the disease, and each degrades without failing the others.

The risk engine combines four public datasets not previously applied to glanders into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. The panel contains 3,078 country-year observations across 162 equine-holding countries from 2005 to 2023, with 172 positive labels and 44 engineered features. Six model families are trained under three validation schemes, and a three-stage chained pipeline is evaluated in which animal population is forecast before occurrence is predicted.

On a forward temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 0.056, roughly a ten-fold lift, with an expected calibration error of 0.016. Performance falls to 0.304 under leave-country-out validation, which is the estimate to quote for an unseen country. Chaining produces no measurable gain in discrimination. An ablation shows climate covariates transfer poorly across countries, and the model performs least well in low-income regions. The dominant limitation is that the label reflects documented activity rather than true incidence, so the model largely learns where disease is reported. Reporting effort is the principal confounder and the main target for future work. The explanation and alert components are specified to the level of their interfaces, rule semantics and evaluation protocol, and are not reported as measured results because they are not deployed.

Keywords: Glanders, *Burkholderia mallei*, Risk Prediction, Gradient Boosting, Probability Calibration, SHAP, Retrieval-Augmented Generation, Citation Grounding, Alert Fatigue, Modular System Design, MLOps, Animal Health Surveillance.

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Chapter 1

Introduction

1. Background

Glanders is a contagious and potentially fatal disease of horses, donkeys and mules caused by the Gram-negative bacterium *Burkholderia mallei*. The organism has no environmental reservoir. Infection persists through transmission among equids, and asymptomatic carriers can remain infectious for extended periods. Human infection is rare but carries a high case fatality rate, and occurs through occupational contact with infected animals. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic significance of glanders is concentrated in low and middle income countries, where equids remain essential to transport, construction, agriculture and brick manufacture. India recorded 5.44 lakh equids in the 2019 Livestock Census, a large proportion held by households that depend on the animals for daily income. For such households the loss of a working animal is a direct loss of livelihood, which makes early warning valuable even where mass treatment is not feasible.

Surveillance data indicate that the disease is neither rare nor geographically contained. A national programme screening 158,798 equid serum samples from 391 districts between 2019 and 2023 identified 728 seropositive animals in 138 districts, with roughly sixty per cent of positives reported from Uttar Pradesh. Historical surveillance covering 2006 to 2018 reported 1,145 positive equids among 128,404 sera tested. These figures describe a persistent, spatially structured problem rather than isolated outbreaks.

2. Problem Definition

The operational question facing a district veterinary officer is not whether glanders exists in the country, but whether it is likely to appear in a given area in the coming year. Answering it requires combining several kinds of evidence: the history of the disease in and around the area, the size of the susceptible population, and the environmental conditions that permit transmission.

Surveillance systems report what has already happened. Forecasting systems exist for several livestock diseases but not for glanders. The problem addressed here is therefore the construction and honest evaluation of a population-level risk model for glanders using only publicly accessible data.

The problem has four parts.

1. **Data assembly.** No single public source records glanders occurrence completely. Occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common geographic and temporal resolution.

2. **Label construction.** The outcome is rare, which makes naive accuracy useless and requires an explicit treatment of what a negative observation means.

3. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.

4. **Threshold economics.** A probability is not an alert. The operating point must be chosen so that the volume of warnings is defensible relative to the events caught.

3. Objectives

The project has one research objective and three engineering objectives. The research objective is the prediction problem; the engineering objectives are the three components that turn the prediction into something a person can use.

1. **Research.** To assemble a country-year panel for glanders from public sources not previously applied to this disease, train and compare six model families under three validation schemes, and report the achievable skill honestly, including negative results.
2. **Component 1, prediction as a service.** To package the trained model as a versioned, portable artefact and expose it through a REST interface, so that the risk engine can be called by any consumer without that consumer knowing how the model works.
3. **Component 2, cited explanation.** To build a retrieval-augmented layer that explains a prediction using official guidance, cites every claim, and refuses rather than guesses when the corpus does not support an answer.
4. **Component 3, owner alerts.** To build a notification service that converts a calibrated probability into a small number of messages, using threshold, suppression and deduplication rules so that a rare disease does not train its users to ignore warnings.

A cross-cutting requirement applies to all four: each component must be independently testable, independently replaceable, and free of any hardcoded reference to glanders, so that the platform can be repointed at another disease without altering the components themselves.

4. Scope and Limitations

The prediction study is scoped to country-year resolution. District-level modelling, which the original project proposal envisaged for India, is not attempted here because the required district-level covariates could not be assembled from public sources within the project period. The explanation and alert layers are specified at area level and are resolution-agnostic, so they apply unchanged once a district-level model exists.

The model is a population-level risk tool. It does not diagnose an individual animal and is not a substitute for laboratory confirmation or clinical examination.

Three limitations are stated at the outset because they shape every result that follows. First, the label is derived from published reports, so a negative observation may indicate absence of disease or absence of reporting. Second, occurrence is reconstructed by an automated extractor that has not been validated against a hand-annotated reference set. Third, with 172 positive observations the statistical uncertainty on every headline metric is substantial, and this is quantified rather than hidden.

5. Organisation of the Report

Chapter 2 reviews the epidemiological and machine learning literature and identifies the gap addressed here. Chapter 3 sets out the system architecture and the three components the platform is built from. Chapter 4 describes the data, the panel construction, the feature engineering and the modelling approach for the risk engine. Chapter 5 covers model serving and the operational layer that turns the trained model into a callable component. Chapter 6 specifies the retrieval-augmented explanation layer and Chapter 7 the alert service. Chapter 8 presents and discusses the evaluation results. Chapter 9 concludes and sets out future work.

Chapter 2

Literature Review and Gap Analysis

1. Glanders Epidemiology and Burden

Glanders is reviewed in the veterinary literature as a neglected zoonosis with high case fatality and a genuine occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates across published studies and identifies management practice and animal movement as the dominant epidemiological drivers [3]. Regional reviews document continued circulation in Brazil [4], and the disease remains endemic or repeatedly reported across parts of Africa, Asia and the Middle East.

The consistent finding across this literature is that glanders persists where equid populations are large, where animals move frequently, and where surveillance is weak. The first two conditions are measurable from public data. The third affects whether the disease is visible at all, and is therefore a confounder rather than a covariate.

2. Surveillance in India

Indian surveillance provides the strongest evidence base for the disease. Malik et al. described the emergence and re-emergence of glanders in India through outbreaks reported between 2006 and 2011 [7]. Singha et al. reported serological surveillance and clinical investigation of indigenous equines for the period 2015 to 2018 [6]. More recently, the national programme covering 2019 to 2023 screened samples from 391 districts and produced a prevalence estimate together with a risk analysis [5].

These studies establish where glanders occurs, which districts carry the burden, and which animal-level factors are associated with seropositivity. None constructs a predictive model, and none uses population or reanalysis covariates as predictors.

3. Geographic and Spatiotemporal Risk Analysis

Descriptive spatial analysis of glanders has been reported. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian glanders data [8]. More broadly, species distribution and ecological niche models have been compared as tools for livestock and disease epidemics [9].

This body of work is descriptive. Cluster detection identifies where cases have already concentrated. It does not forecast which cells will be positive in a future period, and that distinction matters for an early warning application, where the value lies in anticipating rather than confirming.

4. Machine Learning for Animal Disease Risk

Predictive modelling of animal disease risk is established for avian influenza. Kane et al. compared ARIMA and random forest models for the prediction of H5N1 outbreaks [11]. Walsh et al. used machine

learning to target active surveillance for avian influenza viruses [12]. Yoo et al. developed dynamic farm-level risk prediction for highly pathogenic avian influenza in the Republic of Korea [13], and Alshatti et al. produced a risk prediction model for outbreaks in Kuwait [10].

This literature supplies the methodological template used here: assemble occurrence, population and environmental covariates at a common spatial unit, fit tree-based ensembles, and evaluate on held-out periods. The template has been applied to influenza, and not to glanders.

5. Machine Learning and Glanders

The only machine learning work specific to *Burkholderia mallei* that could be located is spectroscopic identification of the organism. Moawad et al. developed a Raman spectroscopic assay in which learned models classify spectra to identify *B. mallei* and related species [14]. Stöckel et al. reported Raman spectroscopic detection and identification of *B. mallei* and *B. pseudomallei* [15].

Both are laboratory diagnostic problems. Neither addresses the geographic or temporal distribution of the disease, and neither produces a risk estimate for a population.

6. Text Mining for Disease Event Extraction

Using published literature as a structured source of occurrence data has precedent in epidemic intelligence. Collier surveyed web-based epidemic intelligence and the role of text mining within it [16], and Lejeune et al. described multilingual event extraction for epidemic detection [17].

The present work applies sentence-level event extraction not as a supplementary signal but as the primary source of the label, for the pragmatic reason that official outbreak records are incomplete for many countries. The consequence, that the label inherits both the strengths and the reporting biases of the literature, is analysed rather than assumed away.

7. Prior Art Audit

To test the novelty claim rather than assert it, a reproducible set of queries was run against Europe PMC. Each row of Table 2.1 is a query count, and the full query strings and retrieved titles are retained alongside the project code.

Table 2.1: Europe PMC hit counts for combinations of glanders with each data source and method used in this study.

Query	**Hits**
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

The decisive cells are the query pairing glanders with NASA POWER, which returns zero records, and the pairing with gradient boosting, which returns four, none of which is a glanders risk model. The nearest prior work is descriptive cluster detection in Brazil and a risk-factor analysis of the Indian serosurvey.

8. Research Gap

The literature reviewed above divides into three disjoint groups. Epidemiological studies describe where glanders occurs. Machine learning studies of *B. mallei* classify laboratory spectra. Veterinary risk models exist but are built for influenza. No study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single panel for glanders, and none evaluates a calibrated predictive model on that panel.

A second gap appears once the question moves from analysis to use. A probability on a dashboard does not reach the person who owns the animals, and a risk score without a stated basis is not something a field officer will act on. The literature that describes glanders occurrence and the literature that builds spectroscopic classifiers both stop short of the delivery problem: how a forecast becomes a cited explanation, and how an explanation becomes a message that an owner receives and trusts. No work connects a glanders risk model to official control guidance and to a notification path with rules that account for alert fatigue.

The contribution of this project therefore has two parts. The first is empirical: a calibrated population-level risk model for glanders, built on data sources not previously applied to the disease and evaluated under a forward temporal split and a country-holdout split. The second is architectural: a decomposition of the delivery problem into three components with fixed contracts, so that the prediction, its explanation and its alert can be built, tested and replaced independently, and so that the same structure carries over to another disease.

Chapter 3

System Architecture

1. What the Platform Has to Do

The platform answers one question and produces three kinds of output from it. The question is whether glanders is likely to be documented in a given area in the next period. The outputs are a number on a map, an explanation in words, and a message to an owner. Each output is consumed by a different person, at a different time, through a different device, and each can fail without the others failing.

That asymmetry is the reason the system is built as three components rather than one application. A district officer looking at a map needs the prediction immediately and the explanation on demand. An owner with a feature phone needs a short message and nothing else. A researcher auditing a decision needs the cited passages months later. Building a single service that serves all three would tie the alert delivery rate to the latency of an LLM call and would make the risk model impossible to test on its own.

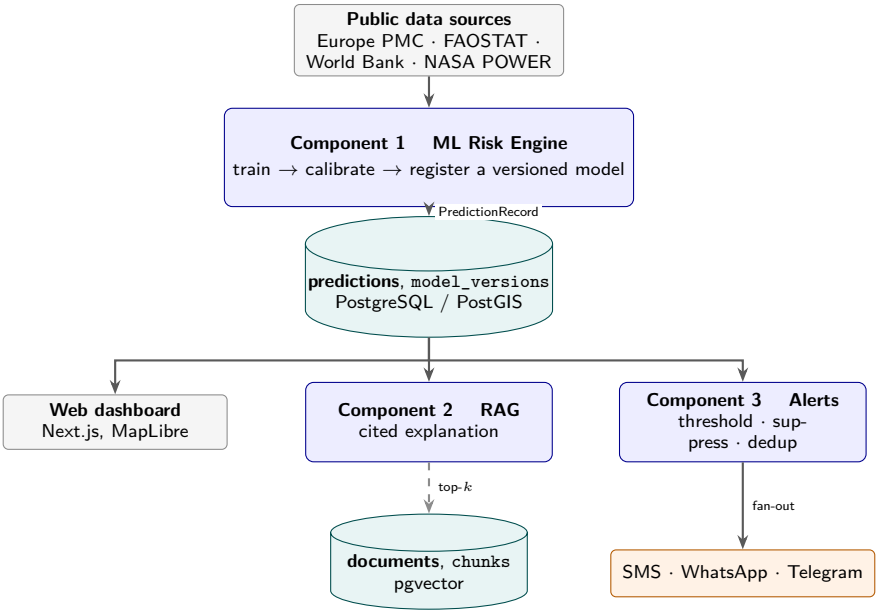


Figure 3.1: The platform decomposed into three components. The ML risk engine writes one `PredictionRecord` per area and period; the dashboard, the RAG layer and the alert service consume it independently. No component reads another component's tables, so any one of them can be replaced without touching the other two.

2. Design Principles

Five rules govern the decomposition. They are stated explicitly because every later design decision follows from them, and because they are the part of this project that transfers to other diseases.

1. One responsibility per component

Each component owns exactly one transformation. The risk engine turns features into a calibrated probability. The RAG layer turns a prediction into a cited explanation. The alert service turns a prediction into a delivered message. None of them computes something another one could compute instead.

2. Contracts before implementations

Components exchange versioned schemas, never shared database tables or in-process calls. The contract is defined first, and each side can be built, tested and deployed against a fixture that satisfies it. This is what allows the RAG layer to be developed while the alert service is still being written.

3. The core is disease-agnostic

Nothing in the risk engine, the RAG layer or the alert service hardcodes glanders. A disease is a configuration record: which label extractor to use, which knowledge-base collection to retrieve from, which message templates to render, which thresholds to apply. Adding a second disease is a configuration change plus a retrain, not a code fork.

4. Fail soft, never fail silent

A degraded component returns a degraded answer, not an error to the user. If the RAG layer is unavailable, the alert is still delivered, with the risk level and the standing action text but without the explanation paragraph. If the explanation is requested and retrieval returns nothing above the relevance floor, the layer returns a refusal plus the top SHAP factors, which are computed by the model and do not depend on the LLM. Neither case produces a wrong statement, and neither produces silence.

5. Every output is traceable

Every prediction carries the model version that produced it. Every explanation carries the document identifiers it was grounded in. Every alert carries the prediction that triggered it and the rule outcomes that allowed it. A decision can therefore be reconstructed after the fact, which is the minimum requirement for a tool that advises on animal disease.

3. The Three Components

Table 3.1: Component responsibilities, interfaces and failure behaviour.

Component	Responsibility	Interface	On failure
ML risk engine	Features to calibrated probability, with attribution	`POST /v1/predict`, `GET /v1/risk-map`	Returns `503` with the champion version id; the last good prediction stays in the table and is marked stale.
RAG layer	Prediction plus question to a cited explanation	`POST /v1/explain`, `POST /v1/chat`	Alert text is sent without the explanation paragraph; the chat endpoint returns a refusal rather than an ungrounded answer.
Alert service	Prediction to a delivered, deduplicated message	`POST /v1/alerts/evaluate`; consumes `predictions_ready`	Alerts queue in Redis and drain on recovery; the map and explanations are unaffected.

4. Component Contracts

A contract here means a JSON schema plus the guarantees that accompany it. The three schemas below are the actual interfaces between the components. They are deliberately small, because a narrow contract is what makes a component replaceable.

PredictionRecord

Emitted by the risk engine, consumed by all three downstream outputs.

```
{
  "schema_version": "1.0",
  "area_id":        "IN-UP-057",        // ISO 3166-2 where available
  "period":         "2026-11",
  "probability":    0.2431,             // calibrated, in [0,1]
  "risk_level":     "HIGH",             // LOW | MEDIUM | HIGH
  "model_id":       "gw-xgb-2026.11.1",
  "predicted_at":   "2026-11-08T00:12:04Z",
  "data_freshness": {"fao_days": 34, "nasa_days": 2, "passes": true},
  "top_factors": [
    {"feature": "equid_density",     "shap":  0.184, "direction": "increases"},
    {"feature": "recent_events_2y",  "shap":  0.121, "direction": "increases"},
    {"feature": "temp_anomaly",      "shap": -0.038, "direction": "decreases"}
  ]
}
```

The guarantees that accompany it are: `probability` is calibrated on held-out data, so it may be read as a frequency; `risk_level` is derived from `probability` by the threshold configuration, so the two cannot contradict each other; and `top_factors` is computed from the model itself, so it is available even when the RAG layer is not.

ExplainRequest and ExplainResponse

```
// request
{ "schema_version": "1.0",
  "prediction": <PredictionRecord>,
  "question":   "Why is this area high risk?",
  "language":   "en" }

// response
{ "schema_version": "1.0",
  "answer":     "Risk is elevated mainly because ...",
  "citations": [ {"doc_id": "dahd-nap-2025", "section": "5.2", "page": 18},
                 {"doc_id": "woah-glanders-2024", "section": "Epidemiology"} ],
  "grounded":   true,
  "refusal":    null,
  "latency_ms": 1840 }
```

The response is grounded when at least one citation resolves to a retrieved passage. A refusal carries `grounded: false`, a reason code and no answer text, so a caller cannot render a refusal as if it were an explanation.

AlertEvent

```
{ "schema_version": "1.0",
  "alert_id":     "a7f3...",          // deterministic, see Section 7.4
  "area_id":      "IN-UP-057",
```

```
"period":       "2026-11",
"level":        "HIGH",
"previous":     "MEDIUM",
"prediction_id": "...",              // foreign key to the prediction
"rules":        {"threshold": "pass", "suppression": "escalation",
                 "deduplication": "new"},
"recipients":   42 }
```

5. Shared Data Model

The components share a database but not tables. Each table has a single owning writer, and the others read it; no component updates another's rows.

Table 3.2: Core tables and their owning writer.

Table	Owner	Key	Read by
`observations`	ingestion job	`(area_id, period)`	risk engine
`features`	feature job	`(area_id, period)`	risk engine
`model_versions`	training job	`model_id`	risk engine, dashboard
`predictions`	risk engine	`(area_id, period, model_id)`	all three
`documents`, `chunks`	RAG ingestion	`(doc_id, chunk_id)`	RAG layer
`subscriptions`	API	`subscription_id`	alert service
`alerts`	alert service	`(area_id, period, level)` unique	dashboard
`alert_delivery_logs`	alert service	`(alert_id, subscription_id)`	dashboard

Spatial columns use PostGIS geometry, and document embeddings use pgvector, so spatial and vector queries run in the same database as the relational joins. This removes a synchronisation problem that a separate vector store would introduce.

6. Implementation Status

The report distinguishes throughout between what has been built and evaluated and what has been specified. Table 3.3 is the honest summary.

Table 3.3: Status of each part of the platform at the time of writing.

Part	Status	Evidence in this report
ML risk engine	Built and evaluated	Panel construction, six model families, three validation schemes, calibration, ablation, SHAP (Chapters 4 and 8). Every number is reproducible from the committed scripts.
Model serving and MLOps	Designed, contract fixed	Endpoint specification, model-bundle format, request lifecycle, monitoring plan (Chapter 5). The model object exists and is serialisable; the HTTP layer is specified rather than deployed.
RAG layer	Designed, contract fixed	Corpus definition, chunking, hybrid retrieval, reranking, grounding and citation checks, evaluation protocol (Chapter 6).
Alert service	Designed, contract fixed	Three rules with their exact semantics, fan-out, delivery logging, worked example (Chapter 7).
Dashboard	Out of scope	Named as the fourth consumer in the architecture; not specified in detail here.

The distinction matters for how the report should be read. The evaluation results in Chapter 8 are measurements. The RAG and alert chapters specify behaviour precisely enough to be implemented and tested, and they state the tests that would validate them, but they do not report measured performance, because the system is not deployed.

7. Modularity and Reuse

The decomposition is only worth its overhead if the parts can be reused. Five concrete reuse paths follow from the design, and each is a small change rather than a rewrite.

Another disease. The risk engine reads its label extractor, feature groups and thresholds from a disease configuration. A second disease supplies a new gazetteer and lexicon for extraction, a new guidance collection for retrieval, and its own tuned thresholds. The modelling code does not change, because it never refers to glanders by name.

Another LLM provider. The RAG layer calls a provider interface with a single completion method, which takes a system instruction, a context block and a question, and a provider adapter supplies that method for either a local model served by Ollama or a hosted API. Switching provider is a configuration value. The retrieval and guardrail code is provider-independent, so the citation check and the refusal path behave identically either way.

Another notification channel. A channel adapter exposes one send operation, which takes a recipient and a rendered message and returns a delivery status. Adding a channel means adding an adapter and a template, not touching the rules engine, which never knows which channel will carry the message.

Embedding the engine as a library. The trained pipeline is a scikit-learn compatible object [32] with `predict_proba`, exported with a manifest that records its feature order, its calibration and its metrics. Any Python process can load it without running the API, which is what makes offline batch scoring and notebook analysis possible.

Replacing the model. Because the prediction record carries `model_id` and the registry keeps previous versions, a new model can be scored against the incumbent on a shadow period before promotion. Nothing downstream changes, since downstream components read the record and not the model.

8. Deployment

The services are containerised and described by a single Docker Compose file: the API, the worker, PostgreSQL with PostGIS and pgvector, Redis, and the frontend. This keeps the local demonstration and a server deployment identical in structure, which matters for a project whose claims are about reproducibility.

The scheduled path is a Celery beat job that runs the feature job, then the prediction job, then publishes `predictions_ready`. Alerts and explanations are computed after that event, so a failed explanation never blocks a prediction and a failed prediction never sends an alert about stale data.

Chapter 4

Methodology

1. Data Sources

Four publicly accessible sources are used: a literature index, a livestock census, a development indicator set and a reanalysis climate archive [38, 35, 37, 36]. None had previously been applied to glanders risk modelling, and each contributes a distinct kind of evidence.

Table 4.1: Data sources and their role in the panel.

Source	Role	Content	Coverage
Europe PMC	Animal history	Occurrence events extracted from abstracts	6,674 records, 263 events
FAOSTAT	Animal population	Stocks of horses, asses, mules, camels, cattle, sheep and goats	68,449 records, 1961–2023
World Bank	Socio-economic context	Ten development indicators and country centroids	217 countries, 2000–2024
NASA POWER	Environment	Temperature, precipitation, humidity and wind	Six measures, 211 centroids, 2005–2024

Figure 4.1 shows the distribution of the extracted occurrence events by year and by country. The volume of recorded activity rises after 2015, which reflects growth in open access veterinary publishing as much as any change in disease incidence. This is one reason the label is treated as documented activity throughout.

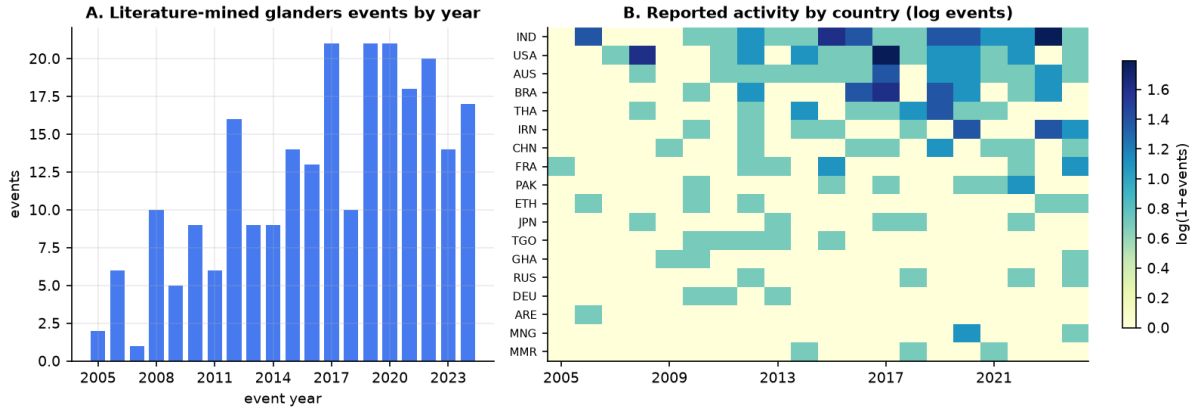


Figure 4.1: Occurrence events extracted from Europe PMC. Panel A gives the annual count. Panel B gives the distribution across the most frequently reporting countries on a logarithmic scale.

2. Panel Construction

All four sources are keyed on ISO 3166 alpha-3 country code and calendar year, so they join without ambiguity once names are reconciled. FAOSTAT is streamed from the 32 MB compressed bulk archive, which expands to approximately 545 MB, and filtered to the seven stock series of interest rather than loaded into memory.

Country names are matched to ISO-3 through `pycountry` with explicit overrides for non-standard forms such as "Türkiye", "Viet Nam" and "Côte d'Ivoire". World Bank regional and global aggregates are excluded by requiring a three-character ISO code, which prevents rows such as "World" or "South Asia" from entering the panel as if they were countries.

The full grid contains 6,700 country-year rows across 268 countries for 2000 to 2024. For modelling, three restrictions are applied: only equine-holding countries are retained, defined as a maximum equid stock of at least one thousand animals; only the years 2005 to 2023 are used; and only rows with an observable next-year label are kept. The resulting matrix has 3,078 rows, 44 features and 172 positive labels.

3. Preprocessing

3.1 Duplicate Removal

Extracted events are de-duplicated on the composite key of country, year and source record identifier. Without this step a single paper reporting several cases would contribute multiple counts and inflate the occurrence signal.

3.2 Missing Values

Missing numeric values are imputed with the column median, and every column that required imputation keeps a binary missing-indicator feature so that absence is represented rather than silently filled. Imputation is fitted inside the training pipeline, so the test period cannot influence imputed values. Missingness is low: 0.00% in the FAOSTAT block, 1.68% in the World Bank block, 2.19% in the climate block, and 0.99% overall, with no column exceeding 5.8%.

3.3 Treatment of Absence

A country-year is assigned a negative label only after the country passes the equine-stock filter. This prevents a country with no recorded equids from contributing an uninformative negative observation, and ensures that a blank is never read as proof that disease was absent.

4. Label Definition

The prediction target is binary: whether a glanders occurrence was documented in country i during year $t + 1$. Occurrence is extracted from Europe PMC by matching, within a single sentence, a country name from a curated gazetteer and a term from an occurrence lexicon containing *outbreak*, *case*, *detected*, *seropositive* and *surveillance*. When both are present, an event is recorded for the year stated in the sentence, or the publication year where no year is stated.

Because the label depends on publication it is termed *documented activity* throughout. A negative value may indicate either the absence of disease or the absence of a report, and the model is therefore interpreted as a predictor of surveillance-observable activity.

5. Feature Engineering

Four groups of features are constructed, all using information available at time t or earlier.

Three derived features require definition. Equid share is the sum of horses, asses and mules divided by total farm animals. Livestock units apply metabolic weights, with cattle and camels at 0.8 and small

Table 4.2: Feature groups, sizes and rationale.

Group	Count	Content and rationale
Disease history	7	Occurrence in the two prior years, a five-year running total, years since the last event, and contemporaneous world and regional activity shares. The disease persists in an area once introduced.
Animal population	14	Log equid and total livestock stocks, equid share of farm animals, equid density, one- and three-year growth, and the seven species counts. A larger susceptible population offers more opportunity for detection.
Climate	10	Mean, maximum and minimum temperature, precipitation total and coefficient of variation, relative humidity and wind, each paired with an anomaly computed against the country's own expanding mean. The anomaly separates unusual conditions from latitude.
Socio-geography	11	Income band, agricultural land share, agricultural value added, livestock production index, GDP per capita, rural share, internet penetration, fertiliser intensity, life expectancy, population and density. These largely proxy surveillance capacity.

ruminants at 0.1, so that one number expresses grazing pressure across species. Years since the last event uses the value 99 to denote no recorded event within the panel window.

6. Feature Selection

Selection proceeded in two stages. First, columns with no measurable association with the label were removed. Mutual information and Spearman correlation were computed for every numeric feature, and columns with mutual information below 0.002 and absolute correlation below 0.02 were discarded. Second, redundancy was addressed. One member of each pair with absolute correlation at or above 0.90 was removed, and a column that was exactly the sum of three others was dropped because a tree model can construct that sum itself.

No linear projection or principal component analysis was applied. With 44 features and 3,078 rows the binding constraint proved to be signal rather than dimensionality. This is supported by the ablation in Section 6., which shows that reduced feature sets generalise better across countries than the full set.

The selection criterion is itself examined in Section 2., where marginal and conditional importance are compared.

7. Models

Six model families are compared. The set spans the range from additive to fully interactive, and includes a deliberately simple reference.

Logistic regression provides an additive baseline with balanced class weights. A single decision tree is included because shallow trees are interpretable by a non-specialist, and because the performance gap between one tree and an ensemble is itself informative. Random forest and extra trees are bagged ensembles [20], differing in the degree of randomisation applied to split selection. XGBoost and LightGBM are boosted ensembles, differing in tree growth strategy [18, 19].

Neural networks were excluded on evidential grounds. With 172 positive observations and 44 features, a network has far more parameters to estimate than it has positive examples from which to learn, and would be expected to memorise the training period and fail on new data.

8. Chained Pipeline

The project proposal specified a multi-stage design. Three stages are implemented.

1. **Population forecast.** Next-year log equid stock is predicted from the current year's covariates

Table 4.3: Model families and the hyperparameters selected by tuning.

Model	Family	Selected configuration
Logistic regression	linear	Additive model with balanced class weights; the reference baseline.
		Tuned settings: $C = 1.0$, ℓ_2 penalty, balanced class weights
Decision tree	tree	Single shallow tree; interpretable, and a low-variance check on ensemble gains.
		Tuned settings: depth 3, minimum leaf size 1, criterion entropy
Random forest	bagged trees	Bagged trees over random subsets; variance control at small positive counts.
		Tuned settings: depth 6, minimum leaf size 1, features per split square root of the column count
Extra trees	bagged trees	Bagged trees with randomised split thresholds; further decorrelation when positives are scarce.
		Tuned settings: depth 6, minimum leaf size 10, features per split 0.3
XGBoost	boosted trees	Boosted trees with explicit regularisation.
		Tuned settings: depth 3, regularisation $\lambda = 1.0$, minimum child weight 5
LightGBM	boosted trees	Leaf-wise boosted trees; efficient comparison on wide tabular data.
		Tuned settings: leaves 15, minimum child samples 40, regularisation $\lambda = 1.0$

using ridge regression and histogram gradient boosting, evaluated by grouped cross-validation on country.

2. **Occurrence.** The binary label is modelled with the forecast population appended as an additional feature, and compared against the same model without it.
3. **Intensity.** Conditional on an event, log event count is modelled by a random forest regressor trained on positive observations only.

The stages are combined into a hurdle model, in which expected event burden is the product of event probability and conditional intensity.

9. Evaluation Design

Three validation schemes are reported, each answering a different question.

- **Temporal holdout.** Training on 2005 to 2017 and testing on 2018 to 2023, with the test period never observed during fitting. This answers whether the model can be used in the following years.
- **Leave-country-out.** Five folds grouped by country, so every prediction is made for a country absent from training. This answers whether the model transfers to a new country, and it governs model selection because it is the harsher test and the closer approximation of deployment.
- **Repeated stratified cross-validation.** Five folds repeated three times, used as a stability check on the ranking metrics.

Precision-recall AUC is the primary metric. At 5.59% prevalence, ROC-AUC is dominated by the true-negative mass and overstates performance, a well-characterised property of imbalanced problems [22, 23]. ROC-AUC is reported for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support a threshold-based alert [25, 26]. Precision, recall, specificity, F1 and the Matthews correlation coefficient are computed at a threshold tuned on training data only, and bootstrap percentile intervals with one thousand resamples quantify sampling uncertainty.

Grouped rather than random splitting is essential. A random row split would place the same country in both training and test partitions, allowing the model to memorise country identity. Given the geographic proxy effects described in Section 2., this would inflate all reported scores [24].

10. Leakage Controls

Six controls are applied.

1. All features are built from year t or earlier, and only the target columns use $t + 1$.
2. The split is chronological, so no future observation can enter training.
3. Hyperparameters are tuned inside training folds using grouped cross-validation.
4. Probability calibration is fitted on training folds only.
5. The alert threshold is fixed on the training period and applied unchanged to test.
6. Missing-indicator features and imputation values are generated inside the pipeline, so the test period cannot influence them.

Chapter 5

MLOps and Model Serving

1. From a Notebook Model to a Service

A trained model is not a component until it can be called, versioned, monitored and replaced. The gap between the two is what this chapter addresses. Three things have to be true before the risk engine can be called a service.

The model has to be a portable artefact, so that the exact object scored in evaluation is the object served in production, with nothing re-fitted in between.

Every prediction has to name the version that produced it, so that a decision can be explained months later and a regression can be traced to a release.

The service has to degrade predictably, so that a bad model release or a missing feature does not produce confident nonsense.

2. Software Layout

The project is a sequence of numbered scripts, each producing artefacts consumed by the next. All intermediate outputs are cached, so any stage can be re-run without repeating the stages before it. This layout is also what makes the pipeline inspectable: a reader can run one stage and check it before trusting the next.

Table 5.1: Processing stages and their outputs.

Script	Stage	Output
`00_novelty_audit`	Prior art audit	Query counts and retrieved titles
`01_fetch_data`	Ingestion	Europe PMC, World Bank and NASA POWER payloads
`01c_extract_faostat`	Ingestion	Seven FAOSTAT stock series
`02_build_panel`	Assembly	Panel, extracted events and audit block
`03_train_evaluate`	Modelling	Metrics, predictions and model comparison
`05_dataset_analysis`	Analysis	Association, redundancy and selection statistics
`04_report`, `06_analysis_figures`	Reporting	Figures and summary tables
`09_make_full_report`	Reporting	This document

3. Training Pipeline

Figure 5.1 shows the training sequence. The split is applied before any model sees data, so the test years cannot influence fitting, tuning, calibration or threshold selection.

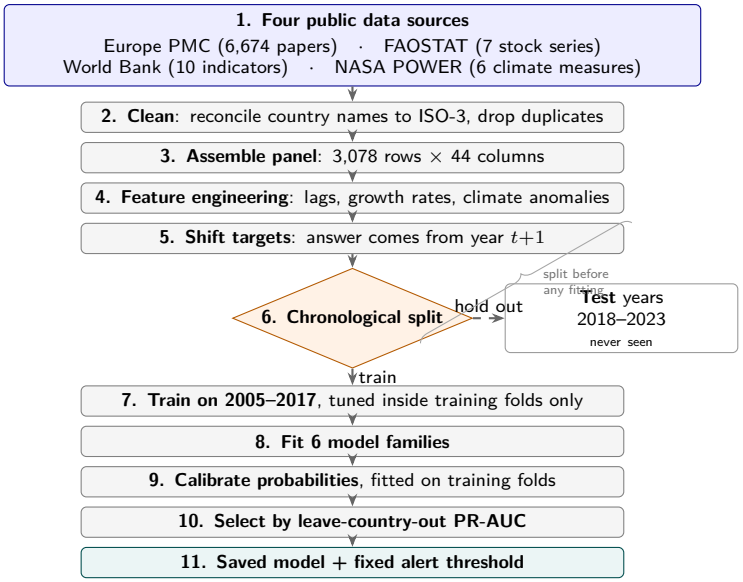


Figure 5.1: Training pipeline. The chronological split sits before any model sees the data, so the 2018–2023 years never influence training. Tuning and calibration both happen strictly inside the training years.

Figure 5.2 shows the corresponding evaluation sequence. One trained model is scored under three schemes, each answering a different question, and the metrics are summarised with bootstrap intervals rather than as point estimates.

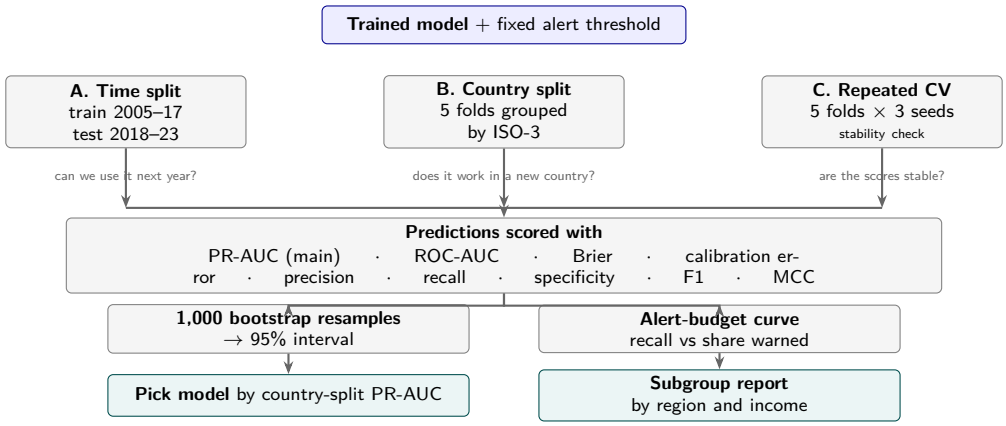


Figure 5.2: Evaluation pipeline. One trained model is scored three different ways, each answering a different question, then reported with uncertainty intervals.

Two properties of this sequence are worth stating because they are commonly violated. Feature scaling and imputation are fitted inside each cross-validation fold, not on the full matrix, so no information leaks from a validation fold into training. The probability calibrator is fitted on a held-out slice of the training period, never on the test period, so the reported calibration error is not circular.

4. Experiment Tracking and the Model Registry

Every training run writes a record containing the git commit, the data version, the feature list and hash, the hyperparameters, and the full metric set for all three validation schemes. Runs are compared on leave-country-out precision-recall AUC, the metric that governs promotion, rather than on the temporal holdout that is more flattering.

A model is promoted from candidate to champion only if it satisfies four conditions:

1. Its leave-country-out PR-AUC exceeds the incumbent's by more than the bootstrap interval width, so the improvement is not noise.

2. Its expected calibration error is below the deployment ceiling, currently 0.05.
3. Its feature schema matches what the serving environment can supply, checked by name and order rather than by position.
4. It has been scored on a shadow period without producing a prediction outside the plausible range.

The registry keeps the previous champion, so promotion is reversible. This is the mechanism behind the fail-soft guarantee in Chapter 3: a bundle that fails its checksum or its schema check is replaced by the previous champion instead of being served.

5. The Model Bundle

The unit of deployment is a directory, not a pickled object, because a pickle alone cannot say what its inputs mean.

```
gw-xgb-2026.11.1/
  model.joblib        # fitted pipeline, predict_proba
  manifest.json       # feature order, dtypes, calibration, metrics
  thresholds.json     # level boundaries and hysteresis margin
  SHA256SUMS
```

The manifest is the part that makes the bundle reusable. It records the ordered feature list with dtypes, so the serving layer can reject a request that supplies the wrong columns before any inference happens. It records the calibration method and the achieved expected calibration error, so a consumer knows whether to read the output as a probability or only as a rank. It records the metrics from evaluation, so the number in the report and the number in the registry are the same number.

Because the bundle is a directory with a manifest, it can be loaded by the API, by a batch job, or by a notebook, with identical behaviour. That is the practical meaning of reusability here.

6. Serving Design

The service exposes a small REST interface. The endpoints are deliberately coarse: a caller asks for a prediction for an area and period, and receives the full record, rather than assembling a prediction from several calls.

Table 5.2: HTTP interface of the risk engine.

Endpoint	Method	Purpose and contract
`/v1/predict`	POST	Score one area and period. Body supplies `area_id`, `period` and either a feature vector or a flag to read stored features. Returns one `PredictionRecord`.
`/v1/predict/batch`	POST	Score up to 500 area-period pairs. Returns a list, with per-item errors rather than failing the request.
`/v1/risk-map`	GET	GeoJSON of the latest predictions for a period, filtered by risk level. Serves the dashboard directly.
`/v1/models`	GET	Registry listing: model id, promotion date, metrics, feature-schema hash.
`/v1/models/{id}`	GET	The manifest of one version, so a consumer can inspect what it is being served.
`/v1/health`	GET	Liveness plus readiness, where readiness checks that a champion bundle is loaded and its feature schema resolves.

Every response carries the `model_id` that produced it and a `data_freshness` block. The freshness block is not decoration: the alert service reads it before sending anything, so a prediction computed on stale upstream data cannot produce a message.

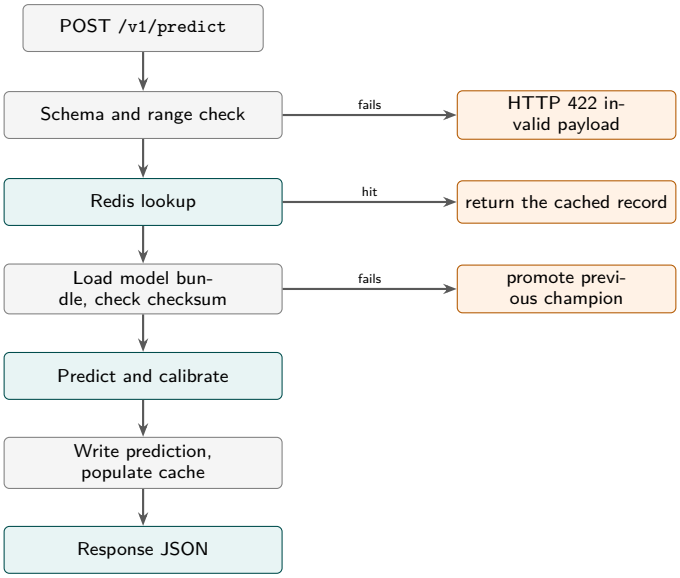


Figure 5.3: Request lifecycle of the prediction endpoint. Validation precedes model loading, the cache is consulted before inference, and a bundle that fails its checksum or its feature-schema check is replaced by the previous champion rather than served.

7. Request Lifecycle

The order of operations in Figure 5.3 is a reliability choice, not an optimisation. Validation runs first so a malformed request costs a schema check rather than a model load. The cache is consulted before inference because risk maps are read far more often than they are recomputed, and the same area-period is requested repeatedly during a dashboard session.

Inference itself is deterministic for a fixed bundle, so a cache key of model id, area and period is safe: the same inputs always produce the same output. The cache is invalidated when a new champion is promoted, which is why the model id is part of the key rather than a version counter.

The response is written back to the predictions table as well as cached, so the read path for the map does not depend on the cache being warm.

8. Latency Budget

The endpoint is specified against a budget rather than measured, since the service is not deployed. The budget is set so that a full risk map for one period is interactive.

Table 5.3: Target latency for a single prediction request.

Stage	Target	Note
Schema and range validation	2 ms	Rejects malformed input before any I/O
Cache lookup	3 ms	Redis, keyed on model, area, period
Feature retrieval	25 ms	Indexed PostGIS and relational joins
Model inference	8 ms	A single row through a tree ensemble
Calibration and record assembly	2 ms	Isotonic lookup plus rounding
Total, cache miss	40 ms	
Total, cache hit	4 ms	

The explanation endpoint has a different budget, two seconds, because it includes a generation step. That is why the alert service does not call it inline: a message must not wait on an LLM.

9. Monitoring

Three classes of signal are monitored, and they catch different failures.

Service health. Request rate, error rate and latency percentiles per endpoint, exposed on a Prometheus endpoint and charted in Grafana. This catches the service being down or slow, which is the least interesting failure.

Data drift. For each feature, the distribution in the current period is compared with the training distribution using a population stability index and a Kolmogorov-Smirnov statistic. A feature that drifts beyond its alerting band is reported, because a tree model asked to extrapolate outside its training range produces a confident answer with no basis.

Prediction drift and calibration decay. The mean predicted probability per period is tracked against the observed event rate as labels arrive. A model that keeps saying 0.2 while the observed rate falls to 0.05 has lost calibration even if its ranking is intact, and the threshold configuration must be revisited before the alerts it drives become wrong.

The retraining trigger follows from the third signal: recalibrate when the observed-to-predicted ratio leaves a band for two consecutive periods, and retrain when a new year of labels is available and the champion's leave-country-out score has degraded beyond the bootstrap interval.

10. Continuous Integration

The pipeline runs linting, unit tests and a smoke test on every push. The unit tests cover the parts that can be checked without data: schema validation, the level-threshold function including hysteresis behaviour at the boundary, the idempotency key derivation, and the citation parser.

A further test checks the model bundle itself. It loads the committed bundle, scores a fixed fixture, and asserts that the output matches a recorded value to four decimals. This catches the most common silent failure in a project of this kind, where a dependency upgrade changes numerical behaviour and every downstream number moves without anyone noticing.

11. Reproducibility

The full computation runs in approximately 4.5 minutes on a laptop processor without a graphics accelerator, using an Apple M4 processor with 10 cores and 16 GB of unified memory. Per-family fitting time at the country-year level ranges from 1.9 seconds for the decision tree to 50.7 seconds for LightGBM.

A single module-level seed governs model fitting, cross-validation splitting, bootstrap resampling and SHAP sampling. Software versions are Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1. Raw payloads are cached under `data/raw` and the derived panel is written to `data/processed`, so every reported number can be regenerated without re-downloading upstream sources.

One determinism caveat applies. Tree ensembles are seeded but not bitwise reproducible across different thread counts, so parallel execution may change the final decimal of an ensemble score. The bundle smoke test described above is run single-threaded for exactly this reason.

Chapter 6

The RAG Explanation Layer

1. The Problem This Layer Solves

A calibrated probability of 0.24 is actionable for a veterinary officer and meaningless for an equine owner. The owner needs to know what happened, what it means for their animals, and what to do next. The officer needs to know which factors drove the number, because a forecast they cannot interrogate is a forecast they will not act on.

The obvious answer is to ask a large language model to explain the prediction. The obvious problem is that a language model asked about glanders will answer from its training data, which may be outdated, may conflate glanders with melioidosis, and may invent a treatment protocol. For a zoonosis with a statutory control programme, an invented instruction is worse than no instruction.

Retrieval-augmented generation [27] resolves this by changing what the model is allowed to use. The model is not asked what it knows about glanders. It is given a small set of passages retrieved from official guidance, given the prediction's own numbers, and asked to answer using only those. Every claim must be traceable to a passage, and the passage must be shown to the reader.

The design goal is therefore narrow and testable: produce an answer that a veterinary officer would accept as consistent with the National Action Plan, and refuse rather than guess when the corpus does not support an answer.

2. Why Retrieval Rather Than Fine-Tuning

Fine-tuning a model on glanders documents would produce fluent text with no citations and no way to tell whether a statement came from the guidance or from the base model's priors. It would also freeze the knowledge at training time, so a revised National Action Plan would require retraining.

Retrieval keeps the knowledge in a corpus that can be updated by re-ingesting a document, makes every answer auditable by construction, and allows a refusal path that fine-tuning does not. The cost is latency and the possibility of retrieval failure, both of which are handled explicitly below.

3. Knowledge Base

The corpus is restricted to sources with authority, because the layer's value is that it cites them. Each document carries an authority tier, which is used as a tie-breaker during retrieval.

Tier 3 material is allowed to inform an explanation but is never allowed to override tier 1 on a control instruction. When two passages conflict, the higher tier wins and the conflict is logged, because a silent contradiction between sources is exactly the failure this layer exists to prevent.

Table 6.1: Knowledge base, by authority tier.

Source	Tier	Use
DAHD National Action Plan for Glanders, 2025 [34]	1	Case definitions, surveillance obligations, movement control, quarantine, notification duties.
WOAH and WHO disease information sheets	1	Disease biology, transmission, zoonotic risk, official status.
ICAR and national laboratory guidance	2	Diagnostic sampling, laboratory referral.
Open-access research papers	3	Epidemiology, risk factors, regional situation.

4. Ingestion and Chunking

Ingestion runs offline as a job, not per request. A document is parsed, split at section boundaries, and each chunk is stored with enough metadata to be cited and filtered.

Chunking is section-aware rather than fixed-width. A fixed 512-token window cuts control instructions in half, so a retrieved chunk can end mid-sentence and the model completes it. The splitter therefore respects heading structure first, then paragraph boundaries, and only falls back to a sliding window for a section longer than the chunk ceiling.

Table 6.2: Chunking parameters.

Parameter	Value
Target chunk size	400 to 700 tokens
Overlap	15 per cent, applied only within a section
Minimum chunk	80 tokens, shorter sections merged into the parent
Metadata retained	doc id, title, section path, page, year, authority tier

The section path is retained in full, for example `NAP-2025 > 5.2 Movement control`, so a citation can point to a specific subsection rather than to a 90-page document.

5. Embedding and Index

Chunks are embedded and stored in pgvector alongside the relational tables. Keeping the vectors in the same database as the predictions removes a synchronisation problem: a re-ingestion and a prediction are covered by the same transaction boundary, and there is no second store to keep consistent.

The default model is a compact sentence-transformer [28] producing 384-dimensional vectors, which is sufficient for a corpus of this size and cheap enough to embed on a laptop. A multilingual variant is used when the corpus or the questions are in Hindi, because an English-only embedding degrades badly on Devanagari text.

The index is HNSW with cosine distance [29]. An approximate index is acceptable here because the reranker corrects recall at the top of the list, and the corpus is small enough that even exact search would be fast; the index is there to keep the query cost flat as the corpus grows.

6. Retrieval

Figure 6.1 shows the complete layer, with ingestion as an offline job along the top and the per-request path below.

Retrieval is hybrid, because dense and lexical search [30] fail in different ways on this corpus. A dense query for "what should I do if I suspect glanders" retrieves passages about suspicion, which is useful. A dense query for the exact phrase "movement control" may miss a passage that uses "restriction of equid movement", where a lexical match would succeed. Conversely a lexical query for a rare acronym retrieves noise that a dense query would ignore.

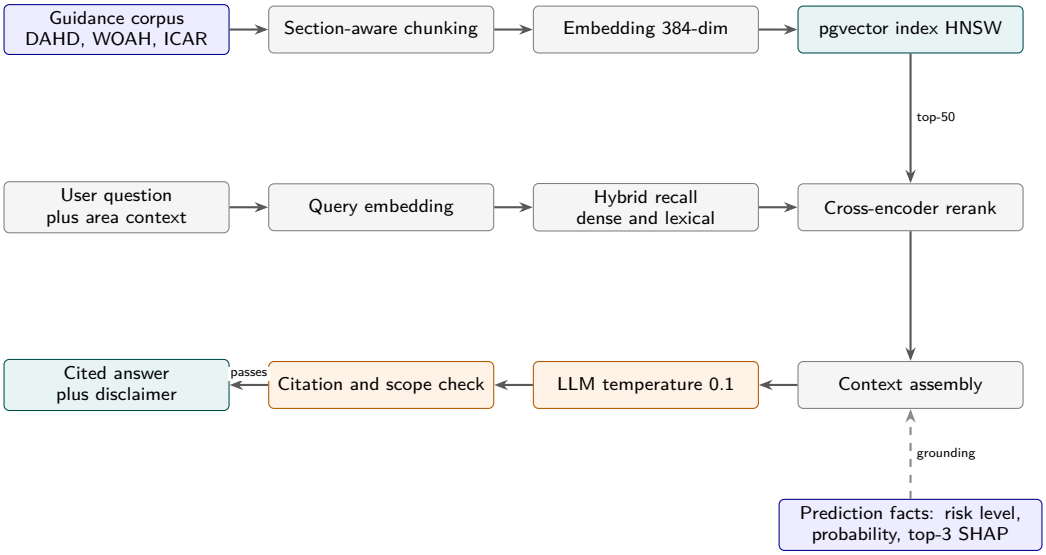


Figure 6.1: The RAG layer. Ingestion (top row) is an offline job; retrieval, assembly and generation (below) run per request. Predicted risk and SHAP factors are injected as structured facts, so the answer explains this forecast rather than glanders in general. The citation check sits after generation and can return a refusal instead of an unsupported answer.

Both result lists are retrieved, then fused by reciprocal rank fusion [31], which combines rankings without requiring the two scores to be on a comparable scale.

$$\mathrm{RRF}(d) = \sum_{i \in \{\text{dense},\text{lexical}\}} \frac{1}{k + \mathrm{rank}_i(d)}, \qquad k = 60$$

The fused top fifty are then reranked by a cross-encoder that scores each passage against the question jointly. A cross-encoder is much slower per pair than an embedding dot product, which is why it runs on fifty candidates rather than on the corpus. The top five surviving passages become the context.

Table 6.3: Retrieval parameters.

Stage	Setting	Reason
Dense recall	top 30, cosine	Semantic paraphrase coverage
Lexical recall	top 30, BM25	Exact terms, acronyms, section titles
Fusion	reciprocal rank, $k = 60$	Scale-free combination of two rankings
Rerank	cross-encoder, top 50 to 5	Precision at the head of the list
Relevance floor	rerank score below 0.30 discarded	Prevents padding the context with weak passages

The relevance floor is the mechanism behind the refusal path. If fewer than two passages clear it, the layer returns a refusal with the SHAP factors instead of an answer, because a context assembled from marginal passages produces a confident answer built on nothing.

7. Context Assembly

The prompt is assembled from three parts, in this order: a fixed system instruction, the retrieved passages with identifiers, and the structured prediction facts.

The prediction facts are what make the answer specific to this forecast rather than to glanders in general. They are injected as structured data rather than as prose, so the model cannot restate a factor as causal when it is only associated.

```
Area:            IN-UP-057, period 2026-11
Risk level:      HIGH
Probability:     0.2431  (calibrated; 4.3x the base rate)
```

```
Model:           gw-xgb-2026.11.1
Top factors:     equid_density      +0.184
                 recent_events_2y   +0.121
                 temp_anomaly       -0.038

Passages:
[P1] dahd-nap-2025, 5.2 Movement control
     ...
[P2] woah-glanders-2024, Epidemiology
     ...
```

The system instruction is fixed and is not influenced by user input. It states four constraints: answer only from the passages; cite every claim with a passage identifier; never diagnose an individual animal; and state that the output is a population-level forecast.

8. Generation

Generation uses temperature 0.1 and a bounded output length. A low temperature is chosen not for creativity but for consistency: two officers asking the same question about the same area should receive the same instruction, and a creative sampling temperature makes that untrue.

The provider sits behind an interface with a single method, so the same retrieval, grounding and citation code runs against a local model served by Ollama or a hosted API. This is the reuse path described in Chapter 3: the guardrails are provider-independent, so switching model cannot silently disable them.

9. Guardrails

The layer is designed to refuse. Four checks run, and any of them can terminate the response.

Table 6.4: Guardrails and their failure behaviour.

Check	What it verifies	On failure
Retrieval floor	At least two passages above the relevance threshold	Refusal with the top SHAP factors, which need no retrieval
Citation resolution	Every cited identifier exists in the retrieved set	Answer rejected and regenerated once; then refusal
Diagnosis scope	No individual-animal diagnosis or treatment prescription	Response rewritten without the clinical claim, and the attempt is logged
Out-of-scope	Question is about glanders control, not unrelated topics	Refusal with a statement of what the layer covers

The citation check is the important one and is mechanical rather than prompted. The answer is parsed for citation identifiers, and each one is looked up in the retrieved set. An identifier that does not resolve means the model cited a document it was not given, which is a hallucination regardless of whether the sentence happens to be true. Prompting the model to be accurate is not a control; checking its output is.

10. Evaluation

The layer is evaluated on four things, none of which is whether the prose reads well.

Retrieval quality. A hand-built set of questions with known relevant passages, scored by recall@5 and mean reciprocal rank. This isolates retrieval from generation: if the right passage is not retrieved, no amount of generation quality helps.

Faithfulness. Whether each claim in an answer is supported by the retrieved context, scored by a judge model and reviewed manually on a sample. This is the metric that catches a model that cites a passage but states something the passage does not say.

Citation precision. The fraction of citations that resolve to a passage that supports the sentence they are attached to. Distinct from resolution, which only checks existence.

Refusal behaviour. The refusal rate on a set of deliberately out-of-scope and underdetermined questions, and the false refusal rate on questions the corpus does answer. A layer that never refuses is unsafe; a layer that always refuses is useless. Both numbers are reported.

The evaluation set is built from the questions the layer is designed to answer, not from generic reading-comprehension data, because the failure modes that matter here are specific to control guidance.

11. Worked Example

The following is the intended shape of a response for a high-risk area, with citations to the corpus described above.

> **Question.** Why is district IN-UP-057 rated HIGH this month?
>
> **Answer.** Risk is elevated mainly because the district has a high equine density and there has been reported glanders activity in the surrounding districts within the last two years. Animals in areas with recent reported activity are at higher risk, and movement of equids is a recognised route of spread [P1]. Owners are advised to avoid unnecessary movement or mixing of equids, to isolate animals showing nasal discharge or skin lesions, and to inform the local veterinary authority rather than attempt treatment themselves [P1, P2]. Glanders is zoonotic, so handlers should use gloves when examining a suspected animal [P2].
>
> This is a population-level forecast for the area. It is not a diagnosis of any individual animal, and suspected cases require laboratory confirmation [P1].

Every sentence in that answer is attributable to a retrieved passage, the control instruction cites the National Action Plan, the zoonotic warning cites the WOAH sheet, and the disclaimer is appended by the layer rather than generated.

12. What Would Be Tested Before Deployment

Because the layer is specified rather than deployed, the validation that would precede deployment is stated explicitly, so the claim is falsifiable.

1. Recall@5 above 0.85 on the hand-built question set.
2. Faithfulness above 0.90 on a manually reviewed sample of 100 answers.
3. Citation precision above 0.95, with every failure inspected by hand.
4. Zero individual-animal diagnoses across the adversarial probe set.
5. Refusal on every question in the out-of-scope set, and a false refusal rate below 10 per cent on the in-scope set.
6. p95 latency under two seconds for the full path on the reference hardware.

These are targets, not results. The distinction is maintained throughout this chapter because the layer has not been run against the corpus at the time of writing.

Chapter 7

The Alert Service

1. Why the Alert Is the Hard Part

The risk engine produces a probability and the RAG layer explains it, but neither reaches the person who owns the animals. For most Indian equid owners the interaction is a message on a phone, not a dashboard, and the message competes with everything else on that phone.

The design problem is therefore not how to send a message. It is how to send few enough messages that the ones which arrive are read. A system that warns about every area every month trains its users to ignore it, and then fails silently at exactly the moment it was built for. Alert fatigue is the dominant failure mode of an early-warning system for a rare disease, and the rules below exist to prevent it.

2. Trigger Path

The service is event-driven. When the prediction job completes it publishes `predictions_ready`, and the alert evaluator consumes it. There is no polling and no scheduled evaluation of its own, which means an alert can never be sent for a period whose predictions were not recomputed.

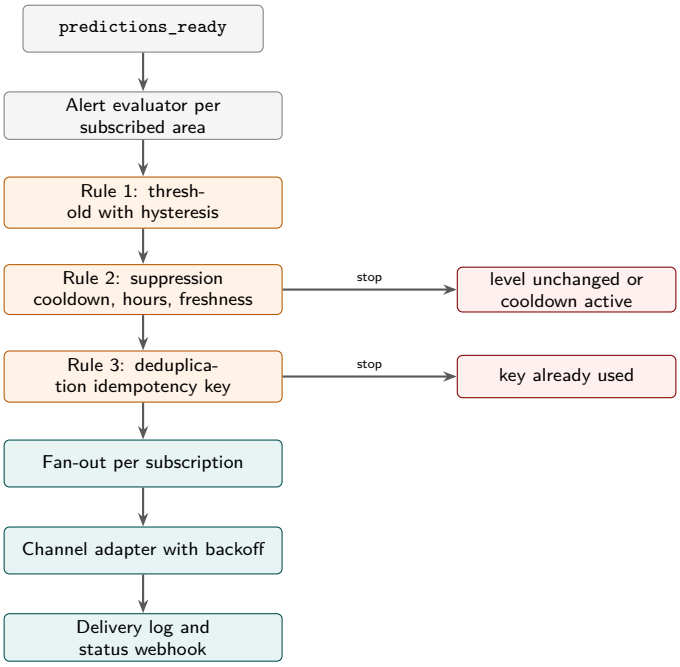


Figure 7.1: The alert rules engine. Each rule can terminate the path on its own, which is what keeps a rare disease from generating routine noise: a level that has not changed and a key that has already been used are both stops, and only an escalation bypasses the cooldown.

For each area with at least one active subscription, the evaluator reads the new prediction and the area's previous alert state. Those two values are the entire input to the rules. The evaluator does not read features or model internals, so a model change cannot alter alert behaviour except through the probability it produces.

3. Rule 1: Threshold with Hysteresis

The calibrated probability is mapped to one of three levels. The boundaries are configuration values derived from the operating curve in Chapter 8, not defaults and not 0.5.

Table 7.1: Level thresholds and their basis.

Level	Band	Meaning for the recipient	Basis
LOW	below the medium boundary	No message. Routine monitoring.	Below the alert budget where precision collapses
MEDIUM	medium to high boundary	Prepare; check animals; note the veterinary contact	Operating point where recall exceeds 80 per cent
HIGH	above the high boundary	Act: restrict movement, isolate suspects, inform the authority	Top of the operating curve, precision above 0.5

Hysteresis addresses a specific defect. With a single boundary, an area whose probability oscillates around the threshold produces an alert every period, because each crossing is a state change. The rule therefore uses two boundaries: an area enters HIGH at the high threshold and only leaves HIGH when the probability falls below that threshold minus a margin.

$$\text{enter HIGH at } p \geq T_H, \qquad \text{leave HIGH only at } p < T_H - \delta, \qquad \delta = 0.05$$

The margin is configurable per disease. A larger margin produces a more stable state at the cost of slower de-escalation, and for a disease where the cost of a missed warning exceeds the cost of a late all-clear, that trade is the right one.

4. Rule 2: Suppression

Suppression decides whether a message is warranted at all, given that the level has changed. Six conditions are applied, and any one of them can stop the message.

Table 7.2: Suppression conditions.

Condition	Rule	Rationale
State change	Send only on a transition, not while the level is held	A monthly reminder at the same level carries no new information
Cooldown	No repeat of the same level within a configurable window	Prevents oscillation at a level boundary from producing repeated messages
Escalation bypass	An upward transition ignores the cooldown	A genuine worsening must never be delayed by a timer
Per-user daily cap	At most one message per recipient per day	Protects against several areas escalating at once
Quiet hours	Nothing sent between 21:00 and 07:00 local time; queued instead	A warning that wakes a household at night is resented, and is not actionable then
Stale-data guard	No send if `data_freshness.passes` is false	An alert about a prediction computed on stale inputs is worse than no alert

A manual mute per area is also supported, so an officer who knows a zone is under active surveillance can silence it without disabling the service.

De-escalation is handled separately: when a level falls, an optional short all-clear is sent, because a recipient who was told to act needs to be told when to stop. Whether that message is sent is a configuration choice, since an all-clear costs a message but prevents a user from maintaining a restriction indefinitely.

5. Rule 3: Deduplication

The service must be safe to re-run. Jobs are retried, deployments restart workers, and a model may be re-scored for the same period. None of those events may produce a second message.

Deduplication is enforced at three levels, each with its own key and its own database constraint.

Table 7.3: Deduplication levels.

Level	Key	What it prevents
Alert	`(area_id, period, level)`	A re-run of the job creating a second alert for the same period
Recipient	`(alert_id, subscription_id)`	One subscriber receiving the same alert through two fan-out passes
Send	`(alert_id, subscription_id, attempt)`	A retry after a timeout delivering a message that had in fact succeeded

The first two are unique constraints in the database, so the protection does not depend on application logic being correct. The alert identifier is derived deterministically from its key rather than generated randomly, so two concurrent workers computing the same alert arrive at the same identifier and one of them loses the insert. A random identifier would let both succeed.

The send-level check reads the delivery log before each attempt. This handles the specific case where a provider accepts a message, the response times out, and the retry would otherwise deliver it twice.

6. Fan-Out and Channels

Once an alert record exists, one delivery job per subscription is queued through Redis. A channel adapter receives a rendered message and a recipient, and returns a delivery status. Adapters exist for SMS, WhatsApp and Telegram, the last of which is used for demonstration because it requires no commercial onboarding.

Retries use exponential backoff with jitter. A permanent failure, such as an invalid number, is not retried; a transient failure, such as a provider timeout, is. The distinction is recorded in the delivery log, because a silent permanent failure means a subscriber who believes they are covered and is not.

Templates are separate from the rules engine. The engine produces a level and a set of facts; the template renderer produces the words, in the recipient's chosen language. English and Hindi are supported, and adding a language adds a template set rather than touching the rules.

7. Worked Example

A HIGH alert as rendered for SMS. The length is bounded because a long message is truncated by the carrier, and a truncated message that loses the action instruction is worse than a short one.

```
GLANDERS WATCH  |  IN-UP-057  |  NOV 2026
Risk: HIGH (was MEDIUM)
Elevated risk for your area. Avoid unnecessary
movement or mixing of equids. Isolate animals with
nasal discharge or skin lesions and inform the local
veterinary authority.
Helpline: <district veterinary contact>
```

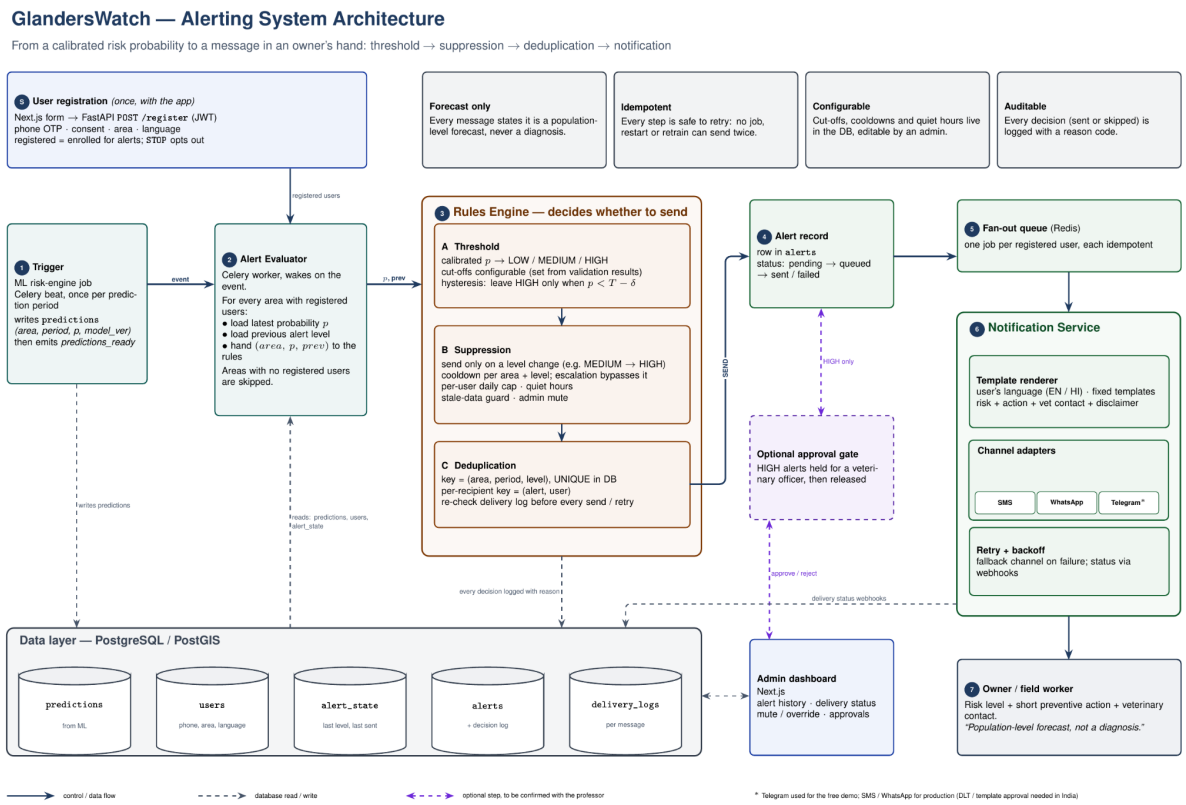


Figure 7.2: The alert pipeline as specified in the project proposal, showing the stages downstream of the rule decision: fan-out to one job per subscription, template rendering, channel adapters, retry with backoff, and delivery status flowing back to the admin dashboard.

`Forecast for the area, not a diagnosis.`

The same alert delivered through a richer channel carries the explanation paragraph from the RAG layer, with its citations. The rule that decides whether a message is sent is identical; only the adapter and the template differ. That separation is what allows one alert to serve a feature phone and a smartphone without duplicating the decision logic.

8. Evaluation of the Alert Layer

The service is evaluated as a warning system, not as a messaging system. Four quantities matter.

Alert precision. Of the area-periods that triggered a HIGH alert, what fraction were followed by a documented event. This is the number that determines whether recipients trust the message.

Event coverage at budget. The fraction of events captured as a function of the number of areas alerted, which is the operating curve already reported in Chapter 8. The thresholds are read off this curve, which is what connects the modelling in this report to the behaviour of the service.

Noise per recipient. Messages per subscriber per year. This is the maintainability metric, because it predicts whether the service will still be read in its second year.

Delivery reliability. The fraction of queued deliveries that reached a terminal success state within the retry window, reported per channel, since channels fail differently.

The first two are computed from historical predictions and labels. The last two require a deployed service and are stated as the protocol that would measure them rather than as results.

Chapter 8

Results and Discussion

1. Dataset Characteristics

The modelling matrix contains 3,078 country-year observations, of which 172 are positive, a prevalence of 5.59% and an imbalance ratio of 16.9 to one. Table 8.1 summarises the distribution.

Table 8.1: Characteristics of the label.

Property	Value
Observations	3,078
Positives and negatives	172 and 2,906
Prevalence	5.59%
Imbalance ratio	16.9 : 1
Countries with at least one positive	56 of 162
Countries positive in two or more years	30
Share of positives in the top five countries	35.5%

Three properties of this distribution matter. The imbalance rules out accuracy as a metric and motivates precision-recall analysis. The temporal persistence of the label, with 30 of 56 positive countries recording positives in more than one year, justifies the use of historical features. The geographic concentration of positives, with the top five countries holding 35.5% of them, provides the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

Figure 8.1 presents the same information graphically, together with the regional distribution of prevalence.

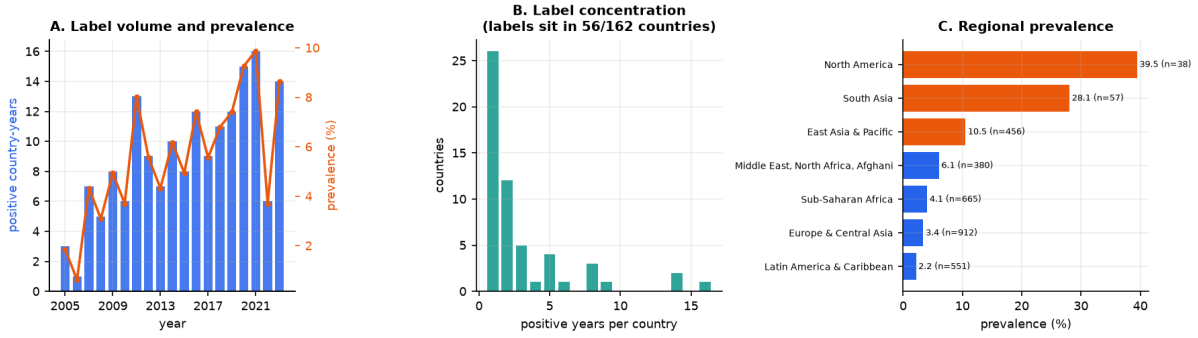


Figure 8.1: Label anatomy. Panel A gives annual positive counts with prevalence on the secondary axis. Panel B gives the distribution of positive years per country. Panel C gives prevalence by World Bank region.

2. Statistical Association and Feature Selection

Spearman correlation, mutual information and Cramér's V were computed for every candidate feature before any model was fitted. The strongest monotone association is `years_since_last` at $\rho = -0.35$, negative because a longer interval since the last recorded event implies lower risk. This is the most defensible epidemiological signal in the panel. Among the categorical strata, region yields a Cramér's V of 0.243, a small association, and income band yields 0.047, which is negligible.

Redundancy is substantial. Variance inflation reaches 228 for `temp_mean`, because annual mean, minimum and maximum temperature are near-duplicates at country resolution. Figure 8.2 shows the correlation structure of the highest-ranked features.

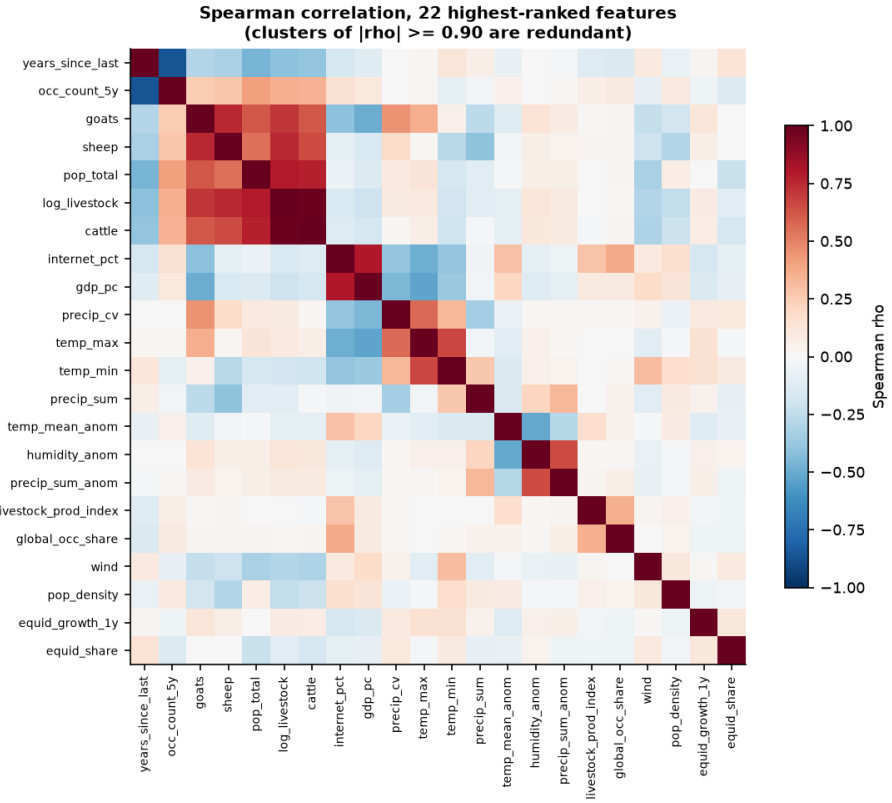


Figure 8.2: Spearman correlation matrix of the twenty-two highest-ranked features, hierarchically ordered. Block structure indicates the collinear groups that motivated redundancy reduction.

The most consequential finding in this section concerns the choice of selection criterion. Marginal association and conditional importance disagree almost completely. The rank correlation between mutual information and SHAP attribution [21] is -0.03, and only 5 of the ten highest-ranked features are shared between the two orderings. `longitude` ranks first on marginal correlation and approximately twentieth on conditional importance, because once population and history are conditioned upon it carries little additional information. It functions as a proxy for which part of the world a country occupies, and therefore for publishing activity.

Figure 8.3 shows the disagreement directly, and Table 8.2 gives the ranked features with both sets of statistics. Ranking features by marginal association alone would favour columns that track reporting practice rather than disease occurrence, so selection was performed on conditional evidence instead.

Figure 8.4 gives the global attribution for the full panel.

3. Model Comparison

Table 8.3 reports discrimination under all three schemes. The temporal holdout is the realistic test and the leave-country-out result is the harder one.

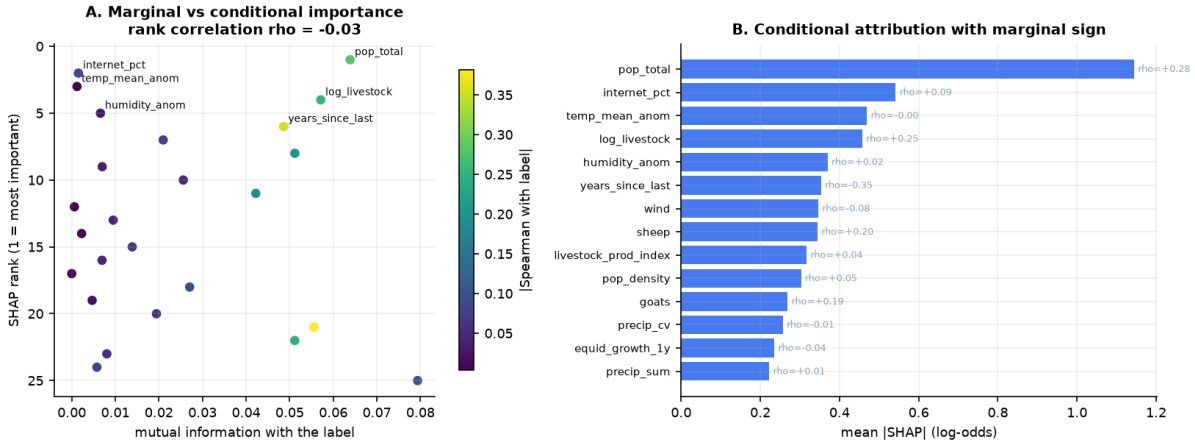


Figure 8.3: Marginal and conditional importance. Panel A plots mutual information against conditional rank, and the absence of a downward trend is the disagreement quantified in the text. Panel B gives the conditional ranking with the marginal association sign annotated.

Table 8.2: The eight highest-ranked features by mean absolute SHAP value, with marginal statistics alongside for comparison.

Rank	Feature	Mean |SHAP|	Spearman ρ	Mutual info	VIF
1	pop_total	1.144	+0.28	0.0639	4
2	internet_pct	0.542	+0.09	0.0015	5
3	temp_mean_anom	0.469	-0.00	0.0012	2
4	log_livestock	0.458	+0.25	0.0571	3
5	humidity_anom	0.370	+0.02	0.0066	2
6	years_since_last	0.353	-0.35	0.0486	2
7	wind	0.346	-0.08	0.0210	3
8	sheep	0.344	+0.20	0.0512	13

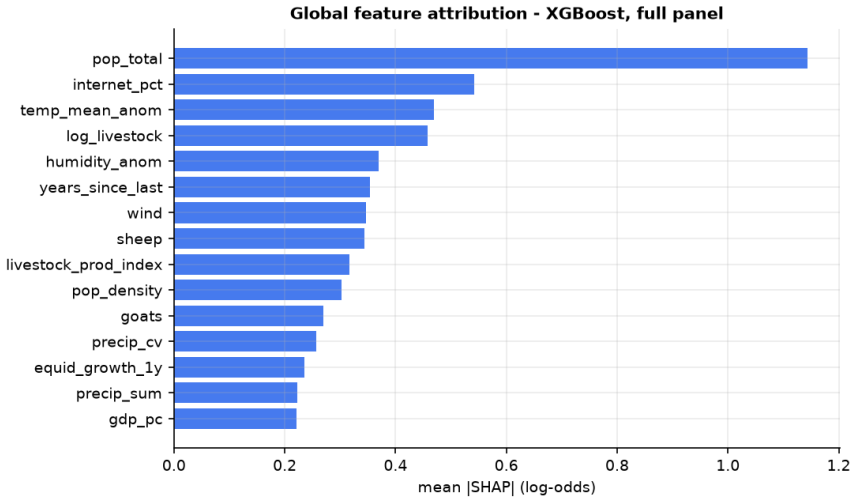


Figure 8.4: Global feature attribution for XGBoost over the full panel, ordered by mean absolute SHAP value. Population magnitude and reporting capacity dominate. Among biological drivers, recent history and climate anomalies rank highest.

Table 8.3: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout.

Model	PR-AUC (time)	95% CI on PR-AUC	ROC-AUC (time)	PR-AUC (country)	PR-AUC (rep. CV)	Brier	Prec.	Rec.
Logistic regression	0.454	[0.34, 0.58]	0.812	0.223	0.414	0.0547	0.25 / 0.57	
Decision tree	0.389	[0.29, 0.50]	0.861	0.267	0.360	0.0602	0.49 / 0.53	
Random forest	0.554	[0.43, 0.66]	0.870	0.304	0.476	0.0503	0.53 / 0.50	
Extra trees	0.530	[0.42, 0.64]	0.854	0.363	0.476	0.0535	0.47 / 0.51	
XGBoost	0.485	[0.37, 0.60]	0.844	0.281	0.423	0.0536	0.61 / 0.38	
LightGBM	0.496	[0.38, 0.61]	0.841	0.339	0.397	0.0534	0.65 / 0.30	

Against a base rate of 5.59%, the strongest temporal result of 0.554 represents an approximately 10-fold lift. Every tree-based model exceeds the logistic baseline, and the single decision tree is the weakest of the six, which indicates that the relationships are interactive rather than additive.

The more important pattern is the systematic decline under leave-country-out validation. For Random Forest the score falls from 0.554 to 0.304, and the decline exceeds the width of the bootstrap interval for every family. This is the expected signature of learning country-specific structure, and it is the estimate that should be quoted for deployment.

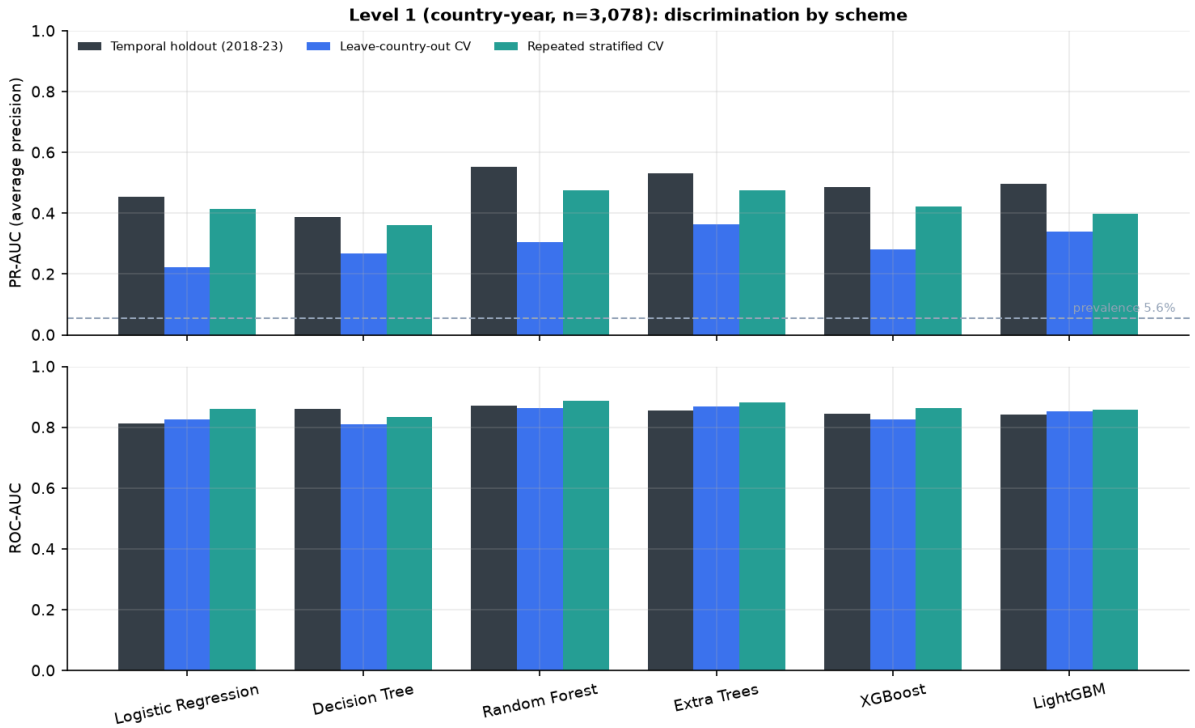


Figure 8.5: Discrimination by model and validation scheme. The gap between repeated cross-validation and leave-country-out validation is the cost of generalising to an unseen country.

Figure 8.6 shows the ROC and precision-recall curves for the temporal holdout. At this prevalence the ROC curves compress into a narrow band while the precision-recall panel separates the families.

4. Calibration

Predictions from the selected model are well calibrated, with an expected calibration error of 0.016 and a Brier score of 0.0535. A stated risk of twenty-four per cent corresponds to an observed event frequency of approximately twenty-four per cent. This property is what makes a threshold-based alert defensible, since a message quoting a probability must be numerically trustworthy.

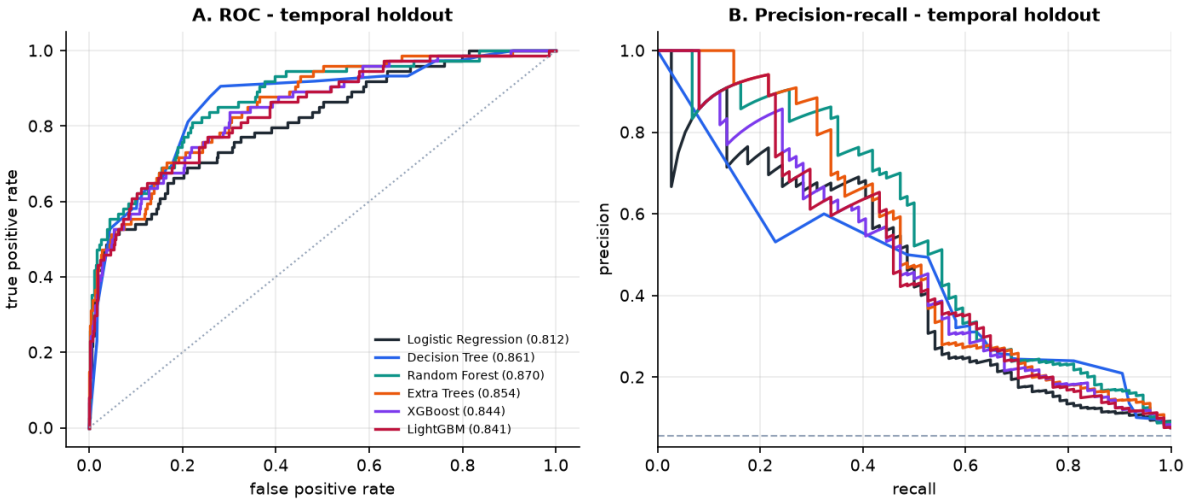


Figure 8.6: ROC curves (panel A) and precision-recall curves (panel B) on the temporal holdout.

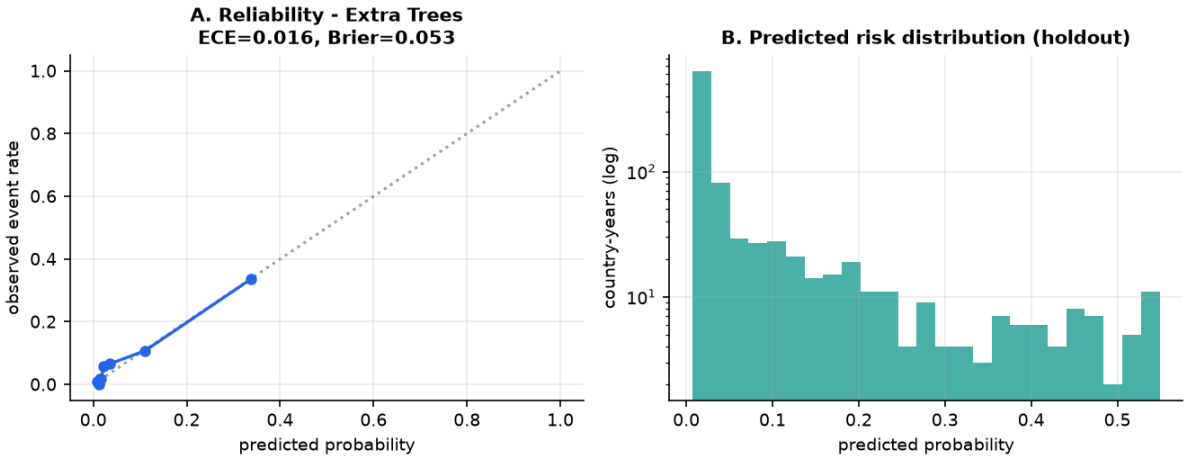


Figure 8.7: Reliability diagram for the selected model on the temporal holdout, together with the distribution of predicted risk.

5. Temporal Resolution

Aggregating observations to country and five-year period raises prevalence to 14.7% and improves every metric, as shown in Figure 8.8. The mechanism is that coarser windows match the reporting cadence of the label, and a positive anywhere within the window counts once.

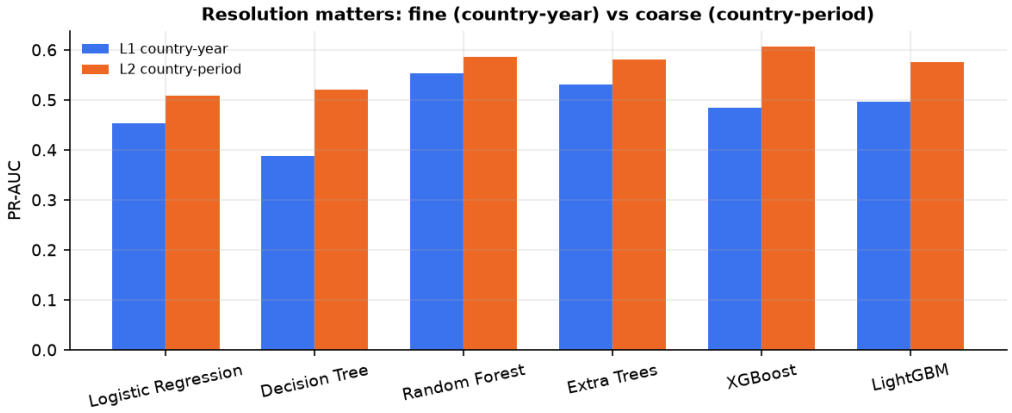


Figure 8.8: The same models evaluated at fine and coarse temporal resolution.

6. Ablation

Feature groups were removed and each configuration retrained, with results in Table 8.4 and Figure 8.9.

Table 8.4: Ablation settings, using XGBoost with fixed hyperparameters.

Feature set	Columns	PR-AUC (time)	PR-AUC (country)
History only	7	0.407	0.174
Population only	14	0.432	0.241
Climate only	10	0.395	0.078
Socio-geography only	11	0.469	0.160
History and population	21	0.451	0.321
All groups	42	0.487	0.259
All groups plus surveillance proxy	44	0.497	0.261

Two results are more informative than the headline score. First, the reduced history and population set of 21 columns reaches 0.321 under leave-country-out validation, against 0.259 for the full 42 column set. Added covariates improve in-distribution fit and reduce transfer, because they allow the model to fit country-specific peculiarities. Second, climate covariates alone reach only 0.078, marginally above the 5.59% floor, which indicates that annual country-level climatology largely encodes latitude and therefore country identity.

A separate probe adds research-output proxies as two additional columns. It moves the leave-country-out score by +0.001, which is weak evidence that the model is not purely tracking publication volume, but it is not proof.

7. Chained Pipeline

Table 8.5 reports the effect of appending the stage-one population forecast to the occurrence model.

The population forecast performs well as a forecast, with a mean absolute error of 0.424 on the log scale and a Spearman correlation of 0.976. The intensity stage is weaker but usable, with a mean absolute error of 0.150 and a rank correlation of 0.465 among positive observations. The hurdle product ranks expected annual event burden with a Spearman correlation of 0.326, and its highest decile captures 70 of 106 observed events.

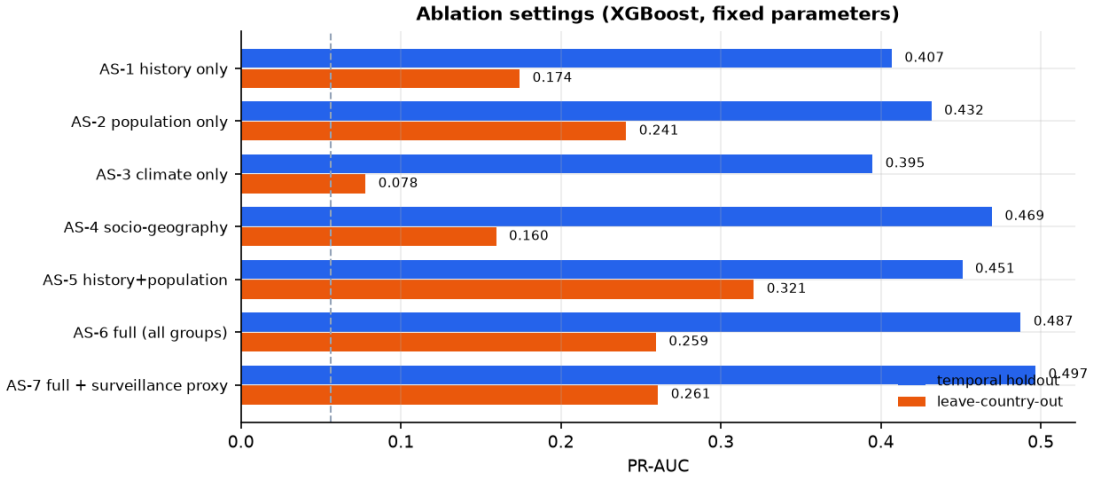


Figure 8.9: Ablation settings under both validation schemes.

Table 8.5: Effect of chaining the population forecast into the occurrence model.

Variant	PR-AUC (time)	ROC-AUC (time)	PR-AUC (country)
Plain (no chained feature)	0.530	0.854	0.363
Chained (population forecast added)	0.527	0.853	0.344

Appending the stage-one forecast to the occurrence model changes temporal PR-AUC by -0.003, which is inside the noise band. The explanation is structural rather than a failure of tuning. Population magnitude is already a direct feature of the occurrence model, so forecasting it and appending the forecast supplies a less accurate copy of information the model already holds. Chaining is expected to assist where the intermediate quantity cannot be observed directly, which is not the case here.

This is reported as a negative result. It is a genuine finding about chained architectures for this problem class, and it is retained rather than removed.

8. Alert Economics

The operating characteristics of the model as a warning system are shown in Figure 8.10. Alerting the six per cent of country-years with the highest predicted risk captures approximately half of all observed events, and alerting twenty-eight per cent captures eighty per cent. Uniform alerting would require full coverage for the same recall, so the model is roughly eight times more efficient than arbitrary selection at the fifty per cent operating point.

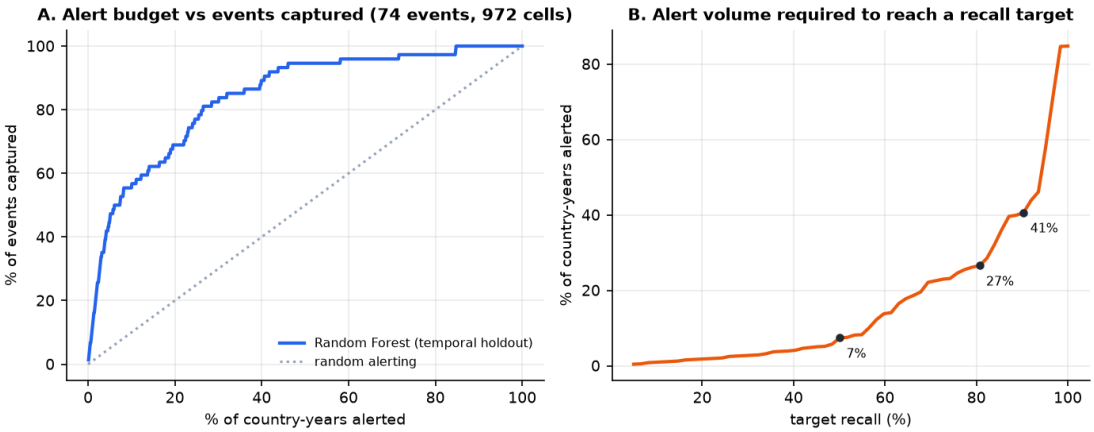


Figure 8.10: Alert economics. Panel A gives the fraction of events captured as the alert budget grows, against random alerting. Panel B gives the alert volume required to reach a target recall.

9. Subgroup Analysis

Table 8.6 reports leave-country-out performance by region. Skill is lowest in Sub-Saharan Africa and in low-income countries, which are the settings in which early warning would be most valuable. The cause is label sparsity rather than a defect in the model, since there are few recorded events from which to learn in those strata.

Table 8.6: Leave-country-out performance by World Bank region.

Region	Observations	Events	ROC-AUC	PR-AUC
East Asia & Pacific	456	48	0.865	0.396
Europe & Central Asia	912	31	0.826	0.163
Latin America & Caribbean	551	12	0.903	0.381
MENA, Afghanistan and Pakistan	380	23	0.834	0.220
Sub-Saharan Africa	665	27	0.699	0.073

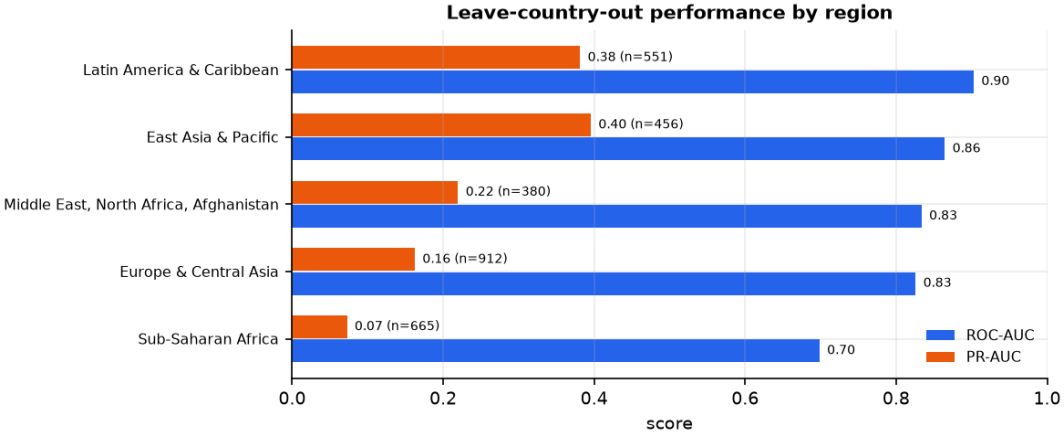


Figure 8.11: Regional performance under leave-country-out validation.

10. Discussion

The results support four conclusions.

The model has genuine predictive skill. A precision-recall AUC of 0.554 against a base rate of 5.59% is not explainable by chance, it survives a strictly forward-looking temporal split, and the probabilities are calibrated to within two percentage points.

The skill does not transfer intact. Leave-country-out performance is roughly forty per cent lower in relative terms, and this is the correct figure for planning. Any deployment to a new country should be preceded by local validation.

The principal signal is surveillance rather than biology. The evidence is consistent across three independent analyses: the highest-ranked features are population and connectivity measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia. Climate, which the original project proposal treated as a primary input, transfers poorly at this resolution.

Simpler models transfer better. The reduced feature set outperforms the full set across countries, and the linear baseline degrades least in absolute terms. This is a practical argument for parsimony when positive examples are limited.

Chapter 9

Conclusion and Future Work

1. Conclusion

This project set out to build a usable warning system for glanders, not only a model. That meant two things: establishing whether population-level risk could be predicted from publicly available data, and designing how such a prediction reaches a person. Four sources that had not previously been applied to the disease were assembled into a country-year panel covering 162 equine-holding countries, and six model families were trained and evaluated under three validation schemes, together with a three-stage chained pipeline. Around that model, the delivery problem was decomposed into three components with fixed contracts: the risk engine that produces a calibrated probability, the retrieval-augmented layer that explains it against official guidance, and the alert service that turns it into a small number of messages.

The central finding is that prediction is possible, and that the achievable performance is moderate. On a strictly forward-looking temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 5.59%, an approximately 10-fold lift, with an expected calibration error of 0.016. Used as a warning system, it captures half of all observed events while alerting six per cent of country-years.

The second finding is that the skill is fragile across geography. Leave-country-out performance is approximately forty per cent lower in relative terms, and the decline is concentrated in the regions where early warning would be most useful.

The third finding concerns what the model is learning. Three independent analyses converge on the same conclusion. The highest-ranked features are population size and reporting capacity rather than biological drivers. The label concentrates in countries that operate reference laboratories. And the highest-scoring cells are the countries with the strongest veterinary publishing record. The model predicts where glanders will be documented, which is useful for directing surveillance, and it is not a map of where the organism is present.

The fourth finding is negative. The chained design specified in the project proposal does not improve discrimination, because the intermediate quantity is already available as a direct feature. This is reported rather than omitted, since it is a reproducible result about chained architectures for this problem class.

The fifth finding is architectural. Decomposing the system into three components with versioned contracts, none of which hardcodes the disease, made each part testable on its own: the risk engine could be evaluated before any explanation existed, and the alert rules could be specified and checked against a policy before any message was sent. The cost is that the parts must be maintained together, and the benefit is that replacing the model, the language model or the notification channel is a local change. For a project whose useful life outlasts any single model, that trade is the right one.

2. Future Work

Six directions follow from the limitations identified in this report. Four concern the model and two concern the components around it.

Modelling reporting effort explicitly. The most important extension is to treat surveillance capacity

as a modelled quantity rather than an incidental covariate. This could be done by using publication volume as a separate offset, or by fitting a two-part model in which detection and occurrence are estimated jointly. Without this, gains in predictive score are likely to reflect reporting practice rather than disease.

Validating the label extractor. The 263 extracted events have not been checked against a hand-annotated reference set. Annotating a stratified sample of several hundred abstracts would allow precision and recall of the extractor itself to be reported, and would bound the label noise that currently propagates into every downstream estimate.

District-level modelling for India. The original proposal envisaged district-month resolution for India, where the disease is concentrated in a small number of states. Public district-level covariates for equine population and climate would need to be assembled, but the resolution would match the scale at which control measures are applied.

Operational calibration under drift. A deployed model requires periodic recalibration as reporting practice changes. The calibration framework used here is a single point estimate; a production system would need drift monitoring and a defined retraining trigger.

Deploying and evaluating the explanation layer. The RAG layer is specified to the level of its retrieval parameters, guardrails and evaluation protocol, but its targets are unmeasured. The next step is to run it against the guidance corpus and report recall@5, faithfulness and citation precision on the hand-built question set, including the false refusal rate, since a layer that refuses too often is as unusable as one that never refuses.

Measuring alert fatigue in the field. The three alert rules are designed to keep message volume low, but the quantity that decides whether the service survives its second year is messages per subscriber per year, and that can only be measured with real subscribers. A small pilot with a district veterinary office would give both that number and the delivery reliability per channel, which the current specification states as a protocol rather than a result.

Two smaller extensions are also worth noting. Incorporating WOAH WAHIS official outbreak records [33] as a second label source would allow the literature-derived label to be validated where the two overlap. And a cost-sensitive threshold, in which the relative cost of a missed outbreak and a false alert is stated explicitly, would replace the F1-optimal threshold used here with one matched to the programme's actual priorities.

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